Scenario 4C: Strong overdispersion stress test with continuous-time x2

MSSTNB posterior performance, dispersion recovery, kappa recovery, and fit-status classification

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1 Purpose and main conclusion

Scenario 4C is a strong overdispersion stress test for the MSSTNB model under the same continuous-time covariate design used in Scenarios 1–3. It preserves the reference mean, exposure, spatial, and temporal latent-risk structure, but reduces the negative-binomial dispersion parameter from $r = 15$ to $r = 3$.

The main result is:

> Under the continuous-time x_2 design, Scenario 4C remains numerically stable in all ten replicates. Strong overdispersion increases count variability and kappa-level variability, but the fitted model preserves accurate regression-slope recovery, learns the smaller dispersion parameter, and recovers the latent temporal-risk path reasonably well.

The formal fixed-gamma sampled-λ analysis produced 10 stable fits, 0 soft-warning fits, and 0 numerical-instability fits out of 10 replicates. There were no beta, kappa, or lambda-output numerical guards.

2 Scenario design

The negative-binomial model is parameterized so that

$$Y_{tj} \sim \mathrm{NB}(\mu_{tj}, r),$$

with approximate variance

$$\mathrm{Var}(Y_{tj}) = \mu_{tj} + \frac{\mu_{tj}^2}{r}.$$

Thus, smaller r means stronger overdispersion. Scenario 4C changes the truth from $r = 15$ to $r = 3$ while retaining the reference dynamic structure and continuous-time x_2 covariate design.

Table 1: Official Scenario 4C continuous-time x2 design.

Quantity	Value
Data scenario ID	S4C_STRONG_OVERDISPERSION_CONTINUOUS_X2_T100
Fit analysis	Fixed-gamma sampled-lambda fit
Stress source	Strong NB overdispersion
Reference dispersion	r = 15
Stress dispersion	r = 3
Ratio to reference	0.20
Exposure	Same as reference dynamic scenario
Mean structure	Same beta/phi/lambda structure as reference dynamic scenario
Time points	100
Coarse areas	9
x2 mode	continuous_time
x2 AR coefficient	0.50
x2 innovation SD	0.80
Fixed gamma	0.80

This design differs from S4A and S4B. S4A creates a global sparse-count regime. S4B creates local information imbalance through low and heterogeneous exposure. S4C instead keeps counts abundant but makes observation-level variability substantially larger through the smaller r.

3 Data calibration

The continuous-time S4C data-generation check passed the calibration criteria. The average count remains high and the zero proportion remains very low, so S4C is not a sparse-count scenario. The stress appears in larger count CV, variance-to-mean ratio, and kappa variability relative to the matched reference $r = 15$ setting.

Table 2: S4C continuous-time x2 data and overdispersion calibration summary.

Quantity	Value
Number of replicates	10
Average count	203.815
Average zero proportion	0.004
Total count	1,834,337
Maximum observed count	4,502
Count CV	1.304
Reference count CV	1.089
Count CV increase	0.214
Variance/mean	353.500
Reference variance/mean	254.214
Variance/mean increase	99.286
Truth kappa CV	0.574
Reference kappa CV	0.257
Kappa CV increase	0.316

The average count is 203.815 and the average zero proportion is only 0.004. The kappa truth CV is 0.574, compared with reference kappa CV 0.257. Thus, S4C successfully creates an overdispersion stress while preserving abundant count information.

4 Fitting strategy

The S4C fit uses the dynamic MSSTNB structure with fixed $\gamma = 0.8$. The model estimates

$$\beta_0, \beta_1, \beta_2, \phi, \tau_\phi, r, \kappa, \lambda,$$

while γ is fixed at its truth. The split component ω is disabled in this analysis. This fixed-gamma strategy isolates strong-overdispersion stress rather than re-testing weak identification of γ.

Numerical guards are recorded and used to classify each replicate. The classification rule is:

- Stable: no beta, kappa, or lambda-output guard and no extreme latent-risk metric.
- Soft warning: small guard activity without severe parameter or latent-process failure.

- Numerical instability: large guard activity, extreme coefficient behavior, extreme lambda RMSE, or extreme log-lambda RMSE.

5 Fit-status classification

Table 3: Fit-status classification for S4C continuous-time x2.

Status	# reps	Percent
Stable	10	100.0

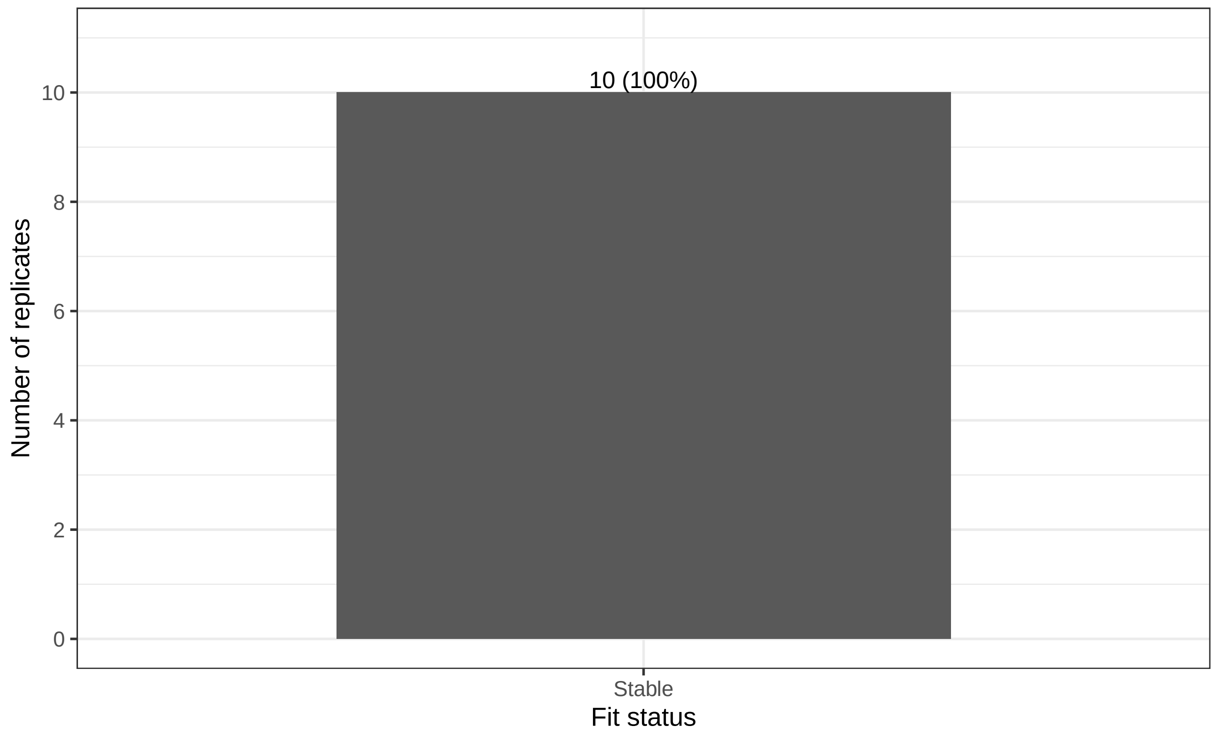

Figure 1: Fit-status counts for S4C continuous-time x2.

All ten formal fits are stable. This is an important difference from the earlier stress scenarios: under the continuous-time x_2 design, strong overdispersion does not create a numerical-failure scenario.

6 Posterior performance

Because all replicates are stable, the all-sampled, stable-only, and stable-plus-soft-warning subsets are identical. The tables below therefore report the main stable set directly.

Table 4: Regression and dispersion recovery for S4C continuous-time x2.

Parameter	Truth	Posterior mean	Across-rep SD	Coverage
beta0	-1.684	-1.689	0.240	1.000
beta1	0.500	0.495	0.022	0.900
beta2	-0.400	-0.405	0.017	1.000
r	3.000	3.194	0.150	0.922

The posterior means for the two slopes are $\bar{\beta}_1 = 0.495$ and $\bar{\beta}_2 = -0.405$, compared with truths $\beta_1 = 0.5$ and $\beta_2 = -0.4$. The empirical coverage is 0.900 for β_1 and 1.000 for β_2. Therefore, strong overdispersion under this continuous-time design does not produce regression-slope fragility.

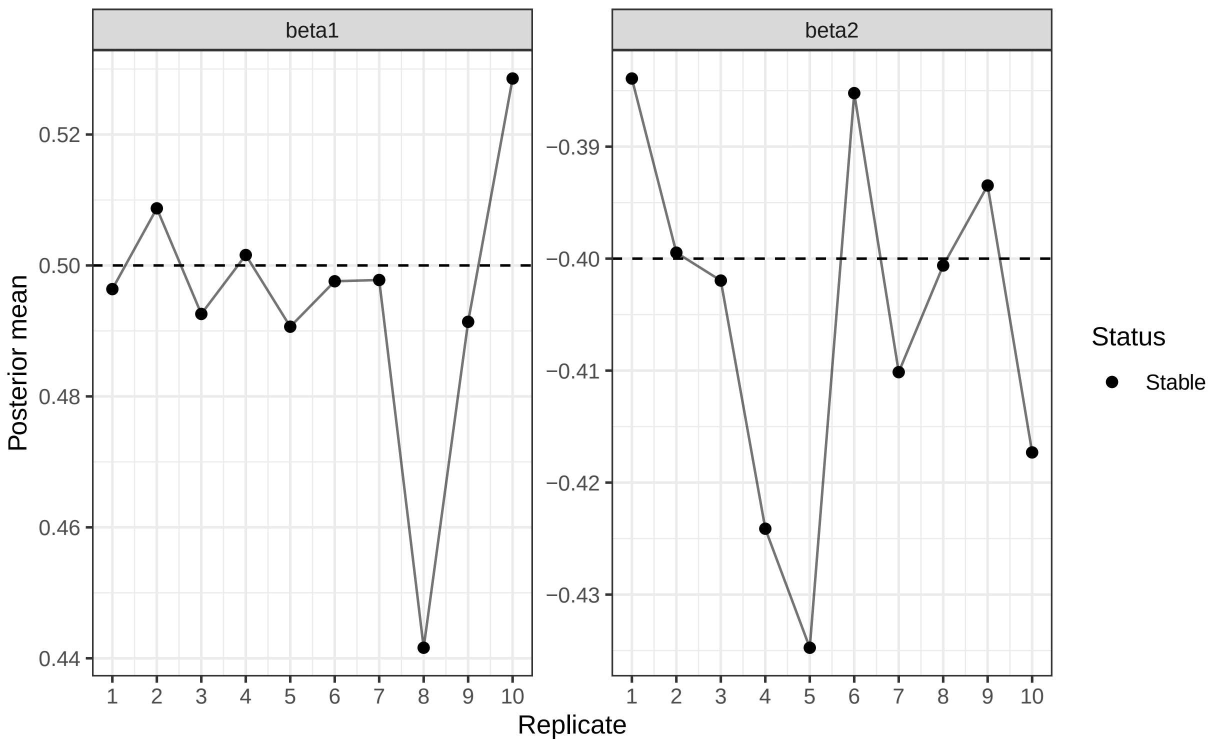

Figure 2: Posterior means of beta1 and beta2 by replicate. Dashed horizontal lines show true values.

7 Replicate-level diagnostics

Table 5: Replicate-level count, regression, and dispersion summaries.

Rep	Status	Mean count	Zero prop.	Count CV	b0	b1	b2	r
1	Stable	151.033	0.020	1.383	-2.150	0.496	-0.384	3.435
2	Stable	226.344	0.000	1.214	-1.407	0.509	-0.399	3.130
3	Stable	218.282	0.000	1.017	-1.432	0.493	-0.402	3.318
4	Stable	132.797	0.018	1.187	-2.027	0.502	-0.424	3.213
5	Stable	297.700	0.000	1.462	-1.576	0.491	-0.435	3.215
6	Stable	218.482	0.001	1.117	-1.612	0.498	-0.385	3.167

Rep	Status	Mean count	Zero prop.	Count CV	b0	b1	b2	r
7	Stable	203.063	0.001	1.501	-1.691	0.498	-0.410	3.210
8	Stable	179.492	0.002	1.645	-1.792	0.442	-0.401	3.305
9	Stable	195.579	0.000	1.247	-1.598	0.491	-0.393	2.887
10	Stable	215.379	0.001	1.262	-1.609	0.529	-0.417	3.062

Table 6: Replicate-level latent-risk and kappa recovery summaries.

Rep	log lambda RMSE	cor(log lambda)	kappa truth CV	kappa post CV	log kappa RMSE	cor(log kappa)
1	0.118	0.832	0.593	0.552	0.253	0.925
2	0.059	0.579	0.562	0.550	0.129	0.980
3	0.053	0.621	0.571	0.550	0.140	0.975
4	0.095	0.892	0.577	0.546	0.217	0.938
5	0.064	0.250	0.568	0.548	0.138	0.976
6	0.077	0.597	0.585	0.564	0.156	0.968
7	0.078	0.753	0.564	0.537	0.197	0.954
8	0.072	0.904	0.568	0.544	0.205	0.943
9	0.060	0.640	0.595	0.582	0.140	0.976
10	0.071	0.796	0.555	0.539	0.137	0.977

Replicate 1 has the largest log-lambda RMSE, but it has no numerical guard and accurate regression recovery. Replicate 8 has the lowest β_1 posterior mean among the ten replicates, but it still has stable latent-risk and kappa recovery and no guard activity. Neither replicate should be interpreted as a numerical failure.

8 Dispersion-parameter recovery

Because S4C is a small-r stress test, recovery of r is a central metric.

Table 7: Dispersion-parameter recovery for S4C continuous-time x2.

Group	N	r truth	r mean	r mean SD	mean bias	mean abs. bias	region coverage
All stable reps	10	3.000	3.194	0.150	0.194	0.217	0.922

The average posterior mean of r is 3.194, compared with truth $r = 3$. The model therefore learns the stronger overdispersion level, although the posterior mean is mildly above the truth on average.

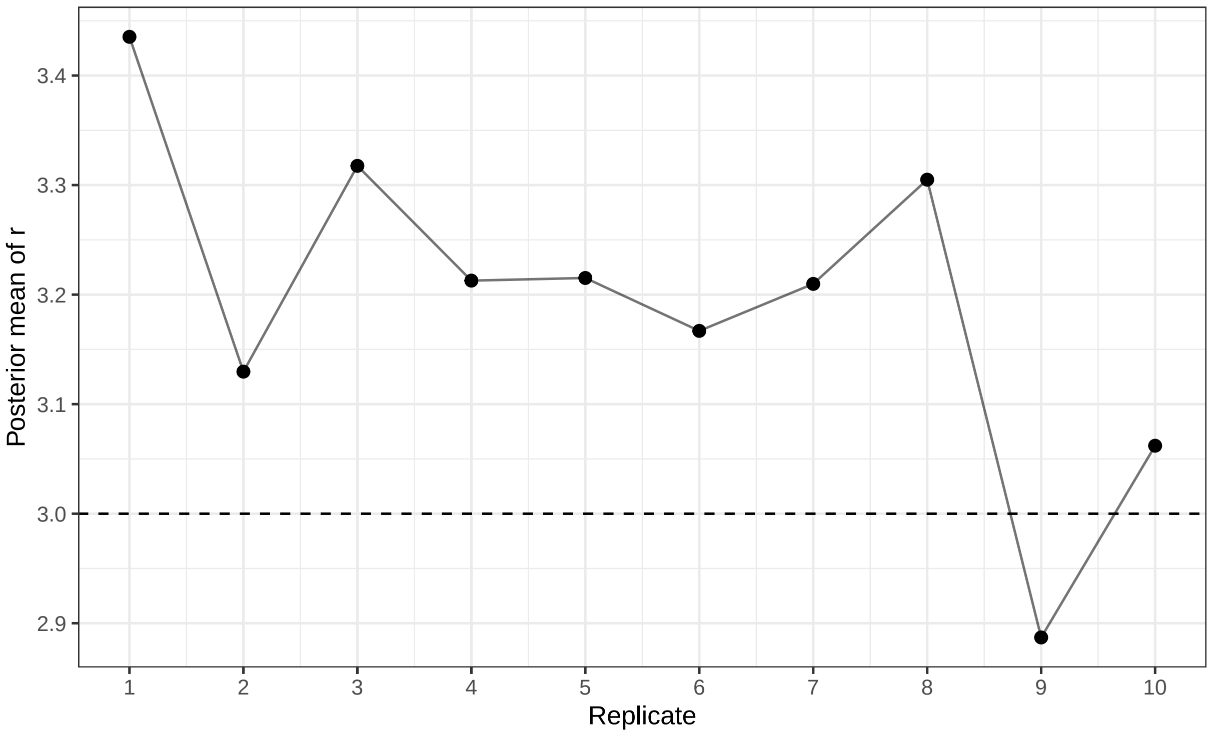

Figure 3: Posterior mean of r by replicate. The dashed line shows the truth r = 3.

9 Kappa recovery

S4C also requires checking the gamma-frailty component κ, because small r increases observation-level overdispersion.

Table 8: Kappa recovery for S4C continuous-time x2.

Group	N	truth CV	reference CV	post mean CV	CV bias	kappa RMSE	log kappa RMSE	cor(log kappa)
All stable reps	10	0.574	0.257	0.551	-0.022	0.156	0.171	0.961

The posterior-mean kappa CV is 0.551, close to the truth CV 0.574 and much larger than the reference CV 0.257. The average correlation for log kappa is 0.961. Thus, kappa-level variability is recovered well.

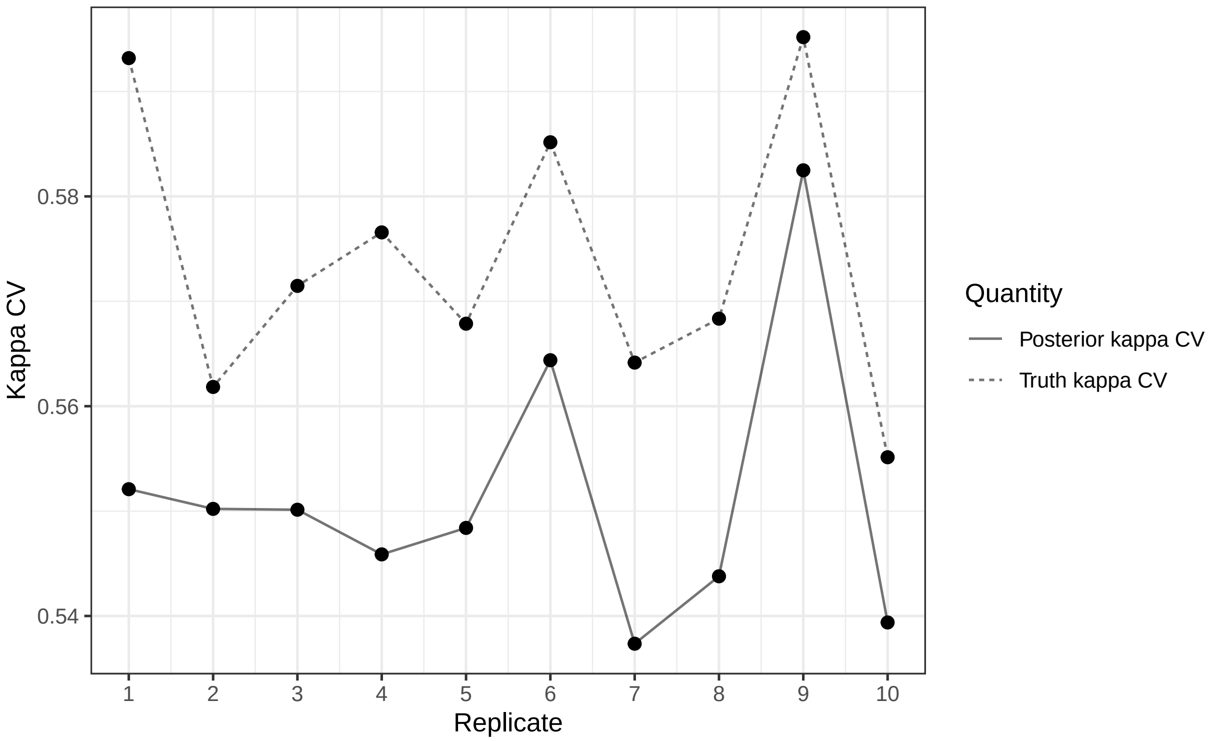

Figure 4: Truth kappa CV and posterior-mean kappa CV by replicate.

10 Latent temporal-risk and spatial recovery

Table 9: Latent temporal-risk and spatial-effect recovery for S4C continuous-time x2.

Group	N	lambda RMSE	log lambda RMSE	median log RMSE	cor(log lambda)	phi RMSE	phi cor
All stable reps	10	0.074	0.075	0.072	0.686	0.050	0.994

The mean log-lambda RMSE is 0.075, and the mean correlation for log lambda is 0.686. This is much stronger than the latent-risk recovery observed in sparse or low-exposure stress regimes and is consistent with S4C retaining abundant count information.

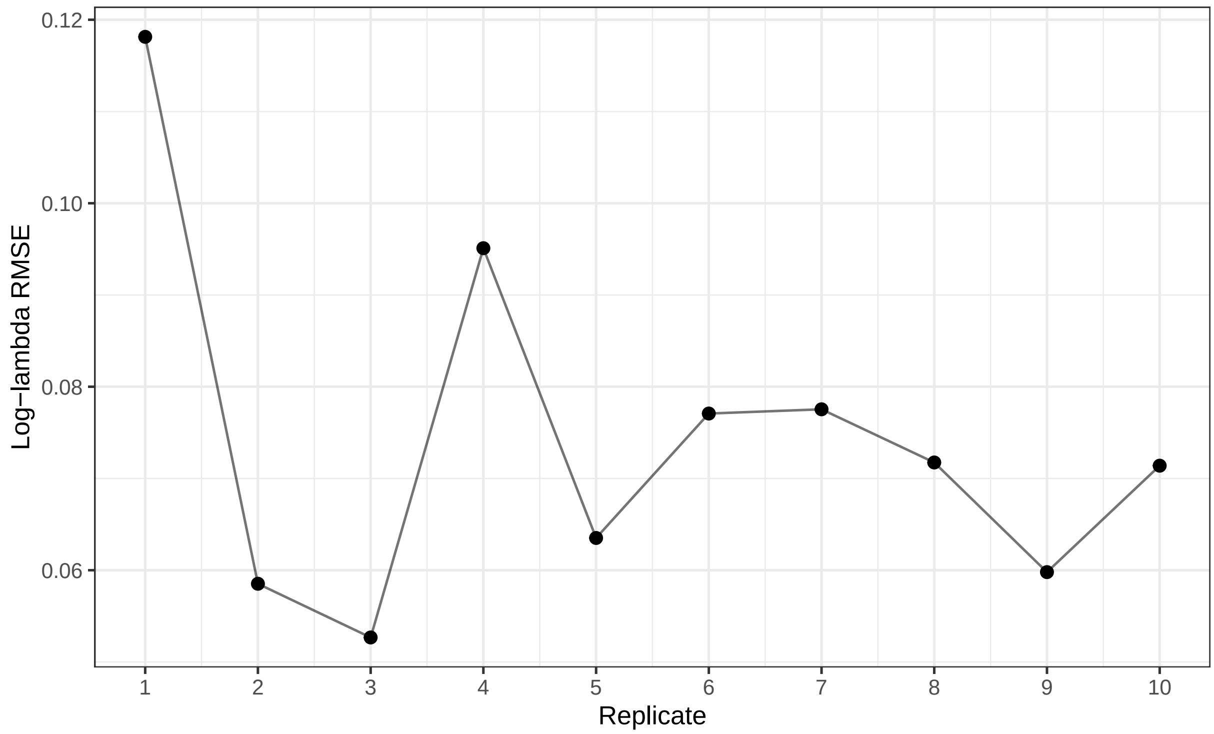

Figure 5: Global log-lambda RMSE by replicate.

11 Numerical guards

Table 10: Numerical guard summary for S4C continuous-time x2.

Group	N	Beta guards	Kappa guards	Lambda-input guards	Lambda-output guards
All stable reps	10	0	0	0	0

There were no beta guards, no kappa guards, no lambda-input guards, and no lambda-output guards. This confirms that the continuous-time S4C result is not a numerical failure scenario.

12 Comparison with the S3 control

Table 11: Comparison of S3 control and S4C continuous-time strong-overdispersion stress.

Scenario	Group	N	Mean count	Zero prop.	b1 mean	b2 mean	r mean	log RMSE	cor(log)
S3 control	control reps	3	195.05	0.000	0.492	-0.408	NA	0.047	0.734
S4C overdispersion	All stable	10	203.82	0.004	0.495	-0.405	3.194	0.075	0.686

Relative to the S3 control, S4C preserves regression-slope recovery while increasing count variability and kappa variability. The S4C log-lambda RMSE is higher than the S3 control but still much smaller than the failure-mode values observed in sparse-count stress settings.

13 Interpretation of the stress mechanism

The continuous-time S4C results separate three phenomena.

First, strong overdispersion increases observation-level variability without creating global sparsity. The average count remains high and the zero proportion remains very low.

Second, the model learns the smaller dispersion parameter and kappa-level variability. The posterior mean of r is close to the truth, and the posterior-mean kappa CV is close to the truth kappa CV.

Third, under the continuous-time x_2 design, regression slopes remain recoverable. This is materially different from a coefficient-fragility or numerical-failure scenario. The dominant S4C message is therefore positive: strong overdispersion is mostly manageable when count information remains abundant.

14 Practical implications

The S4C continuous-time x_2 results suggest several practical implications for MSSTNB applications.

1. Strong overdispersion should be diagnosed separately from sparse counts and low exposure. It increases variability, but it does not necessarily destroy information when counts remain abundant.
2. Dispersion recovery should be explicitly reported. In this setting, the posterior mean of r is close to the truth $r = 3$.
3. Kappa diagnostics should distinguish variability-level recovery from exact cell-level recovery. Here, kappa CV and log-kappa correlation are both informative.
4. Fit-status classification remains useful, but in this continuous-time S4C setting it mainly documents stability.
5. Real applications with strong overdispersion may still need careful posterior checks for r, κ, and λ, even if regression coefficients look accurate.

15 Final conclusion

Scenario 4C with continuous-time x_2 confirms that reducing the negative-binomial dispersion parameter from $r = 15$ to $r = 3$ creates a strong-overdispersion stress regime without creating global sparsity. The data have high average counts, low zero proportions, increased count CV, increased variance-to-mean ratio, and substantially higher kappa variability than the reference setting.

Under this regime, all ten fixed-gamma sampled-λ fits are stable and no numerical guards occur. The regression slopes are accurately recovered: the posterior means are 0.495 for β_1 and -0.405 for β_2, compared with truths 0.5 and -0.4. The model also recovers the stronger overdispersion level, with average posterior mean $r = 3.194$, and recovers kappa-level variability, with posterior-mean kappa CV 0.551 compared with truth CV 0.574.

Therefore, S4C should be interpreted as a strong-overdispersion stress result with successful recovery under abundant count information, not as a regression-slope failure or a beta/r/kappa implementation failure.

16 Recommended reporting language

Scenario 4C evaluated the MSSTNB model under strong negative-binomial overdispersion while using the same continuous-time x_2 covariate design as the main dynamic simulations. The dispersion parameter was reduced from $r = 15$ to $r = 3$, increasing count variability and kappa-level variability without creating a sparse-count regime. In the fixed-gamma sampled-λ fit, all ten replicates were stable and no numerical guards occurred. Both regression slopes were accurately recovered, the posterior mean of r was close to the truth, and kappa-level variability was recovered well. These results indicate that strong overdispersion is largely manageable for the MSSTNB model when count information remains abundant, although latent-risk and frailty diagnostics should still be reported explicitly.